Aluminium-Water Combustion for Zero Carbon Marine Shipping: Masters Proposal

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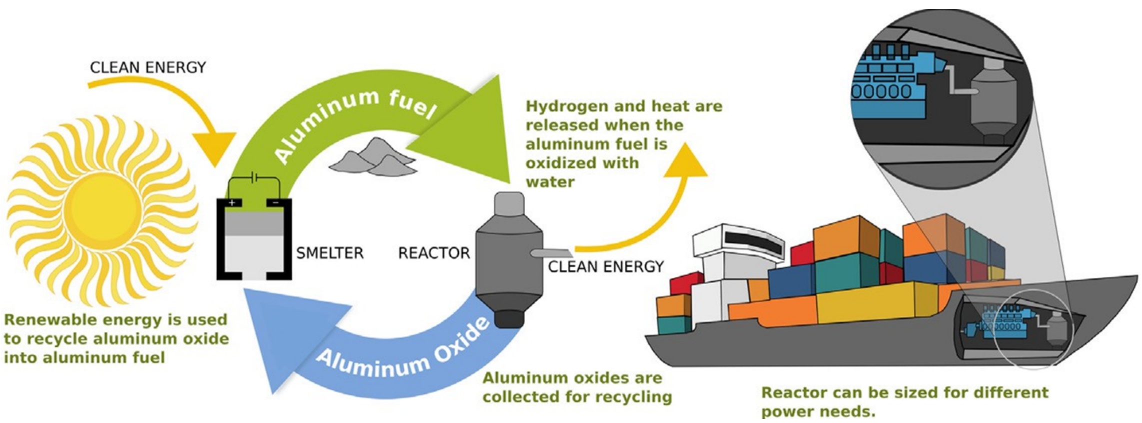

Figure 1: Aluminium powder as maritime shipping fuel (Energy Life-cycle). Adapted from [1]

Abstract

Table of Contents

List of Tables

List of Figures

1 Research Background

Overview

Maritime shipping plays a crucial role in global trade, facilitating around 80% of all commerce [19]. This makes it essential for globalization, economic development, and maintaining living standards. However, the sector also contributes significantly to global greenhouse gas emissions, accounting for 2-3% of global emissions and approximately 10% of all transport-related emissions [20, 19].

In response, the International Maritime Organization (IMO) has set ambitious targets to curb emissions from international shipping. By 2030, the IMO aims to reduce annual greenhouse gas emissions by 20-30% compared to 2008 levels, with a more aggressive target of 70-80% reduction by 2040 [21].

1.1 Preliminary Analysis of Electro-Fuels Options

Overview

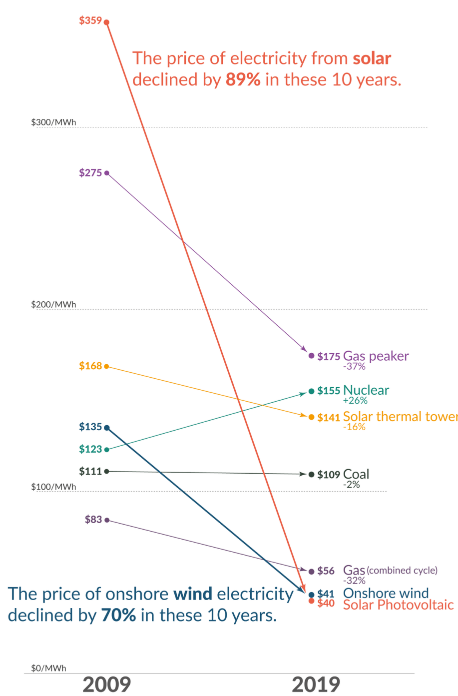

Figure 2: Falling cost of solar electricity vs Fossil Fuels [12]

Recent years have witnessed a dramatic decline in the cost of low-carbon electricity sources, particularly solar and wind power, as illustrated in Figure 2. This cost reduction has significantly enhanced the viability of transitioning to a low-carbon economy, yet substantial challenges persist in the so-called hard-to-abate sectors, among which marine shipping features prominently.

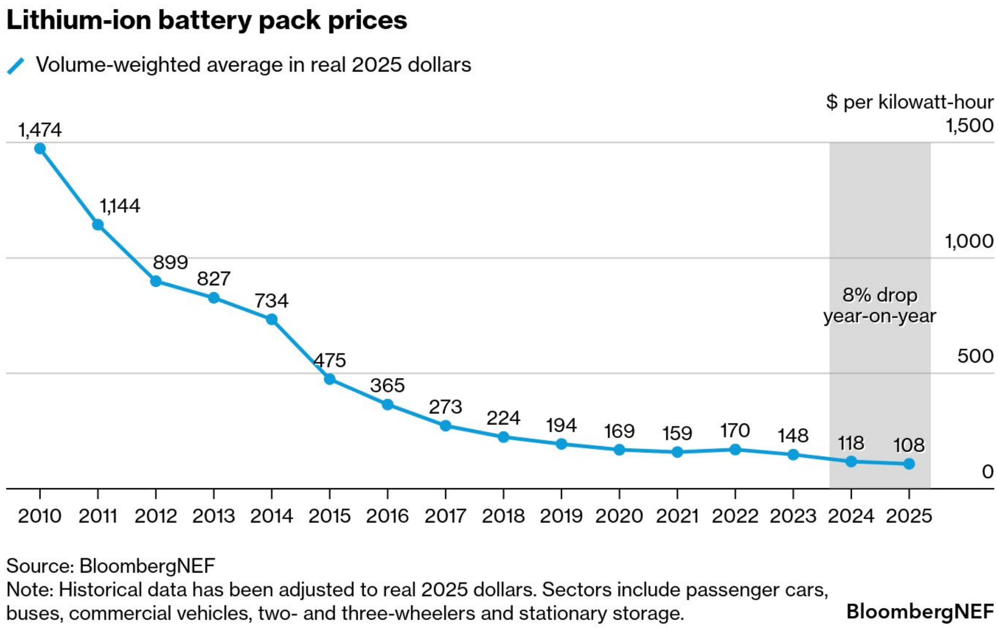

Figure 3: Falling cost of lithium ion battery packs [13]

The predominant method for storing electrical energy remains electrochemical batteries, with lithium-ion technology leading the charge. These batteries have increasingly displaced fossil fuels in passenger vehicles and, more recently, have begun penetrating short and medium-distance trucking applications [22]. This transformation has been driven largely by the precipitous decline in lithium battery prices, as demonstrated in Figure 3. The technology's reach has even extended to inland shipping routes, where standardized and swappable battery packs are being deployed in China [23]. Major battery manufacturers such as CATL are making substantial investments in these markets, signaling confidence in the technology's continued expansion [24].

However, despite remarkable advances in both price and energy density, lithium-ion batteries remain largely impractical for deep-sea shipping, which accounts for approximately 70% of marine shipping emissions [25]. The fundamental constraint lies in the current energy density limitations of lithium-ion technology, which ranges from 0.1 to 0.5 kWh/kg [2, 3]—substantially lower than gasoline, which can exceed 10 kWh/kg [6]. This considerable disparity in energy density presents a formidable barrier to the electrification of long-distance maritime transport.

In response to the energy density limitations of lithium-ion batteries, researchers and policymakers have increasingly explored the use of abundant and inexpensive solar electricity to produce alternative fuels, primarily through electrolysis. This approach has given rise to a diverse landscape of electro-fuel options, which can be systematically organized into four broad classifications:

- **Simple electro-fuels** - hydrogen and ammonia, which represent the most straightforward products of electrolytic processes

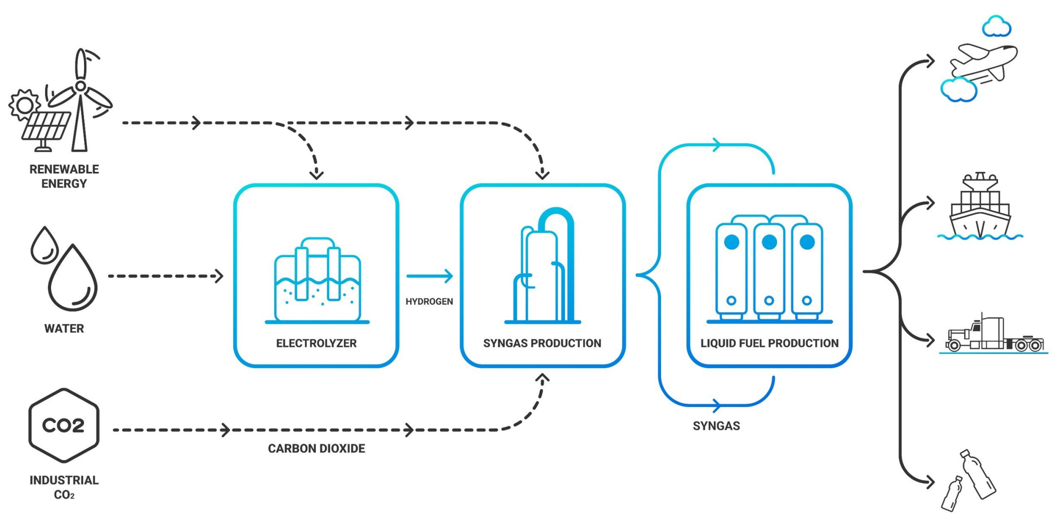

Figure 4: High Level Process Diagram for synthetic hydrocarbon production from renewable electricity [14]

- **Synthetic hydrocarbons** - primarily methane and methanol, whose production pathways are illustrated in Figure 4, which depicts the high-level process for synthesizing these fuels from low-carbon electricity

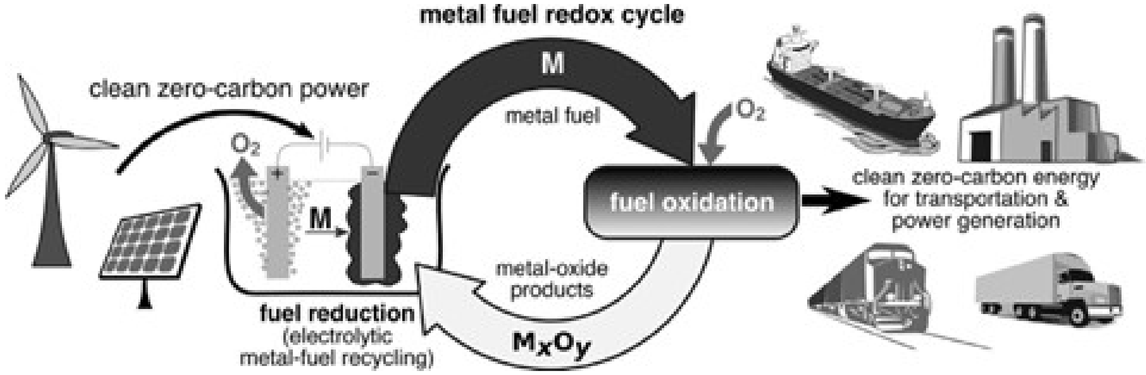

Figure 5: Metal based electro-fuels as secondary energy carriers [15]

- **Metal-based electro-fuels** - including iron, aluminium, magnesium, silicon, and boron. Figure 5 demonstrates the high-level process by which these metal-based electro-fuels can function as recyclable secondary energy carriers, offering a distinctive closed-loop energy storage paradigm
- **Metal hydrides (hydrogen carriers)** - such as MgH_2, AlH_3, $LiBH_4$, and $NaAlH_4$, providing an alternative means of storing and transporting hydrogen in a more dense and stable form than its gaseous state

In the following subsections, a preliminary analysis is conducted to compare the mass energy and volumetric energy densities of these various electro-fuel options in order to understand their viability for marine shipping applications.

1.1.1 High Level Energy Density Analysis

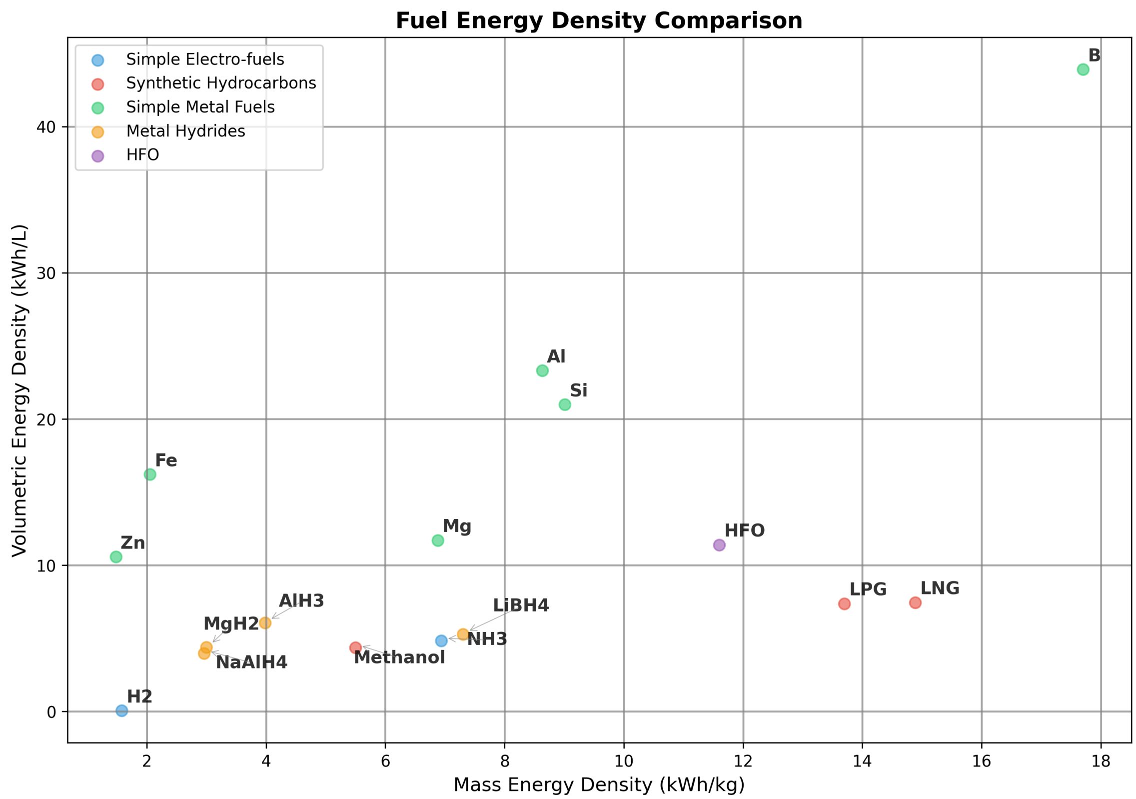

Figure 6: Simplified Energy Density Analysis. Data Analysis can be found in the Appendix. Adapted from [16]

Note: 1) Pressure Cylinder H2 is 25°Celsius 500 bar
2) Liquid NH3 is -33°Celsius 1 bar. Energy needed for cracking for NH3 to H2 is ignored in this table and all subsequent calculations.

Figure 6 presents the fuel energy density calculations for the various electro-fuel options currently under investigation for marine applications. Heavy Fuel Oil (HFO) serves as the reference fuel in these comparisons, as it remains the dominant fuel source for global marine shipping due to its low cost and widespread availability.

Hydrogen, which is being explored as a potential solution for several hard-to-decarbonise sectors including shipping and aviation, requires pressurisation at approximately 500 bar. This compression requirement drastically diminishes the fuel's energy density, presenting significant challenges for practical implementation. Even methanol, which is considered a relatively straightforward drop-in replacement for marine shipping and has garnered substantial support from major shipping companies such as Maersk [26], exhibits significantly lower mass and volumetric energy densities compared to HFO.

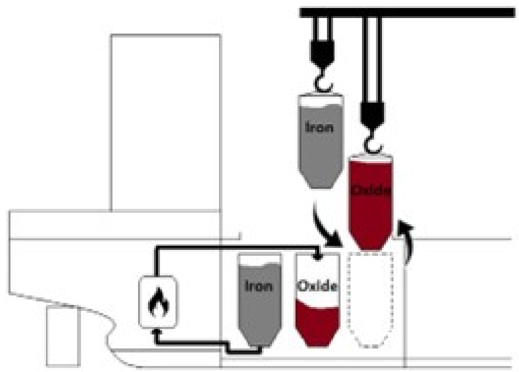

Figure 7: Proposed bunkering procedure for metal fuels (iron powder in this specific case) [17]

As illustrated in Figure 6, metal fuels demonstrate considerable promise in terms of both mass and volumetric energy densities, despite the relatively limited research effort devoted to their development. All metal fuels examined display higher volumetric energy densities than HFO, with boron demonstrating superior performance across all available electro-fuel options and surpassing even HFO itself. However, metal fuels present numerous complications that must be addressed for practical deployment. To make metal fuels economically viable, the metal oxide by-products generated during combustion cannot simply be discharged into the ocean. Instead, these by-products must be retained onboard the vessel and offloaded at ports for subsequent recycling using renewable energy sources, as shown in Fig 5. The bunkering procedure for a metal fuels system is depicted in Figure 7. It is crucial to recognise that energy density calculations which fail to account for the mass and volume of the metal oxide by-products can be highly misleading and do not reflect the true practical performance of these fuel systems.

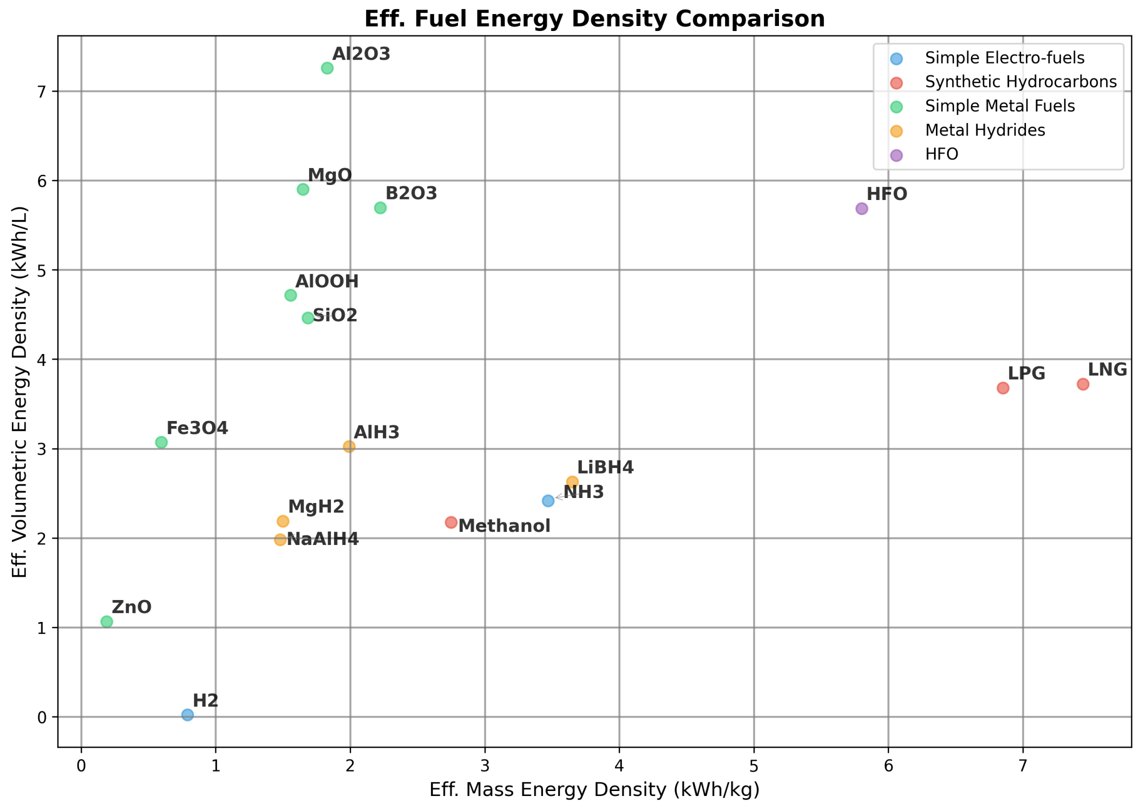

Figure 8: Effective Energy Density Analysis. Data Analysis can be found in the Appendix. Adapted from [16]

Note: 1) Pressure Cylinder H2 is 25°Celsius 500 bar
2) Liquid NH3 is -33°Celsius 1 bar. Energy needed for cracking for NH3 to H2 is ignored in this table and all subsequent calculations.
3) The engine efficiency for hydrogen and hydrocarbon combustion is assumed to be around 50%
4) The engine efficiency for the metal fuels is assumed to be 40%. This number is theortically as no such system currently exists for marine shipping. More information about the metal combustion will be discussed in the following sections.
5) For the metal hydrides, the energy needed to free the stored hydrogen is not considered in this calculation.

1.1.2 Effective Energy Density Analysis

To gain a more comprehensive understanding of the practicality of the different electro-fuel options, an analysis of effective energy densities must be conducted. Effective energy density is defined as the energy density after accounting for engine efficiency, but in the case of metal fuels, it additionally incorporates the weight and volume associated with carrying the metal oxide by-products post-combustion. Figure 8 presents the results from the effective energy density analysis for the various electro-fuel options available. For metal fuels, the effective energy density represents the worst-case scenario of continuously carrying the by-products post-combustion throughout the voyage. In practice, the effective energy density for metal fuels will likely be higher than these calculations suggest, since the by-products can be partially offloaded at ports for collection and subsequent processing at metal reduction facilities.

It should be noted that this is a highly simplified analysis, as the mass and volume of the engines and metal reactors have been deliberately excluded from consideration. Furthermore, there is currently no real-world commercially available marine engine that utilises metal fuels, and therefore an efficiency of approximately 40% has been assumed for these calculations. The complete methodology and detailed calculations for the effective energy density analysis can be found in the Appendix.

As demonstrated in Figure 8, metal fuels remain competitive with the energy density of HFO even after accounting for the metal oxide by-products, particularly with respect to volumetric energy density. Although the mass energy density of the most promising metal fuel candidates, such as aluminium and silicon, remains lower than that of methanol, it is still substantially higher than commercially available electrochemical batteries. The superior volumetric energy densities are of particular significance, given that in practice the metal fuel system will necessitate two sets of storage tanks—one for storing the fuel and another for storing the by-products. This characteristic is also likely to prove advantageous for marine applications where space is at a premium, such as yachts, cruise ships, and container shipping. In later sections, the viability of using aluminium as a marine fuel is discussed in detail, given its promising performance following the effective energy density calculations.

1.2 Aluminium-Water Combustion

Using the energy density analysis from the previous sub-section, aluminium metal emerges as the most promising candidate for use in marine applications. Although it cannot match HFO and other fossil fuels in terms of mass energy density, it demonstrates competitive performance with respect to volumetric energy density. The higher volumetric energy density of aluminium fuel is particularly advantageous for marine applications where space is at a premium, such as yachts, cruise ships, and container ships. Additionally, in practice, the vessel will utilise two storage tanks—one for storing the aluminium powder fuel and another for storing the aluminium oxide by-product, which can be collected later at ports or refuelling stations for recycling.

Aluminium offers several other significant advantages that make it an attractive fuel candidate. It is the third most abundant metal in the Earth's crust, ensuring long-term availability and resource security [1, 27]. Furthermore, aluminium production is already a massive global industry, as the metal's properties make it exceptionally useful across numerous applications. There exists a well-established electrochemical process for producing aluminium from aluminium oxide, known as the Hall-Héroult process. This industrial process can be rendered entirely zero-carbon if the electricity is sourced from renewable energy and the electrolysers make use of inert anodes, thereby eliminating carbon emissions from the production cycle [28].

$$2\,\mathrm{Al} + 3\,\mathrm{H_2O} \longrightarrow \mathrm{Al_2O_3} + 3\,\mathrm{H_2} + 4.21\,\mathrm{kWh/kg{-}Al} \tag{1}$$

$$3\,\mathrm{H_2} + \frac{3}{2}\,\mathrm{O_2} \longrightarrow 3\,\mathrm{H_2O} + 4.40\,\mathrm{kWh/kg{-}Al} \tag{2}$$

Equation 1 demonstrates how aluminium reacts with water to produce heat and hydrogen gas. Equation 2 subsequently shows how this hydrogen gas can then be combusted to produce water and additional heat. Together, these equations reveal that to utilise the full chemical energy potential of the aluminium-water system, both the heat generated from the metal-water reaction and the heat produced from the hydrogen combustion must be harnessed. This metal-water reaction, also known as wet cycle combustion, has already found application in several specialized domains, including ALICE rockets developed for NASA [29] and underwater propulsion applications employed by the American and Russian militaries [30].

1.3 Reactor Design

Overview

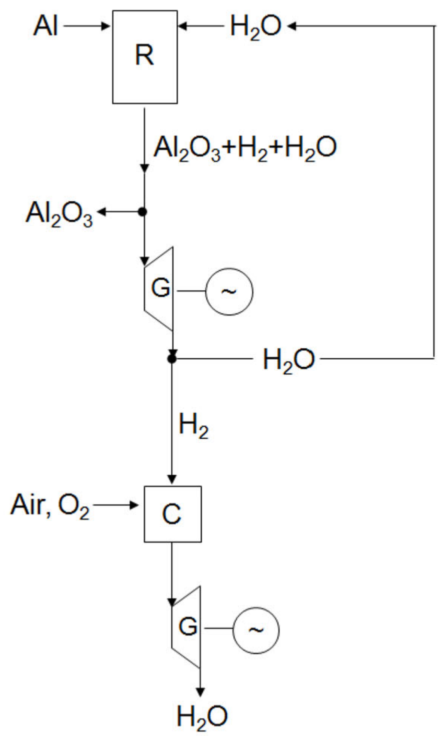

Figure 9: Simplified Overview of Metal-Water Propulsion System[4]

Note: R = metal-water reactor, G = steam generator, C = hydrogen combustion chamber

The metal-water propulsion system, as illustrated in Fig 9, demonstrates a novel approach to maritime propulsion. In this system, seawater from the ocean is mixed with aluminium powder from the fuel tank within a specialized metal-water reactor. The resulting reaction generates three key outputs: hydrogen, thermal heat, and aluminium oxide by-product, with the latter being separated into a dedicated storage tank. The system harnesses energy through two pathways: the thermal heat is utilized to drive a steam engine, while the produced hydrogen can be combusted in a separate chamber to generate high-temperature steam for powering an additional steam engine.

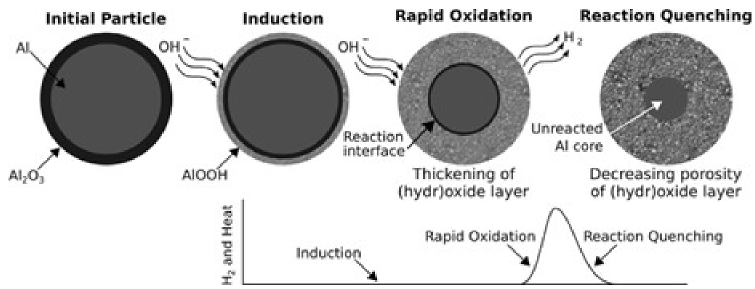

Figure 10: Schematic of the three reaction stages of aluminium with water at low temperatures, including a sketch of the initial dry particle for reference. Also sketched is the characteristic hydrogen-production curve showing the induction, rapid oxidation, and quenching stages [6]

Fig 10 illustrates the reaction pathway for the low temperature aluminium-water reaction, depicting an aluminium particle encased by a thin, inert aluminium oxide coating that must be overcome to facilitate metal-water oxidation. The metal-water reactor design requires careful optimization to achieve appropriate reaction rates for marine propulsion applications, while simultaneously maximizing hydrogen yield to enhance both the efficiency and mass energy density of the aluminium fuel.

The reaction rate and hydrogen yield can be enhanced through:

- Increased temperature and/or pressure conditions
- Reduced aluminium powder particle size
- Reduced aluminium powder to water ratio
- Implementation of various catalysts (detailed in following sub-section) [6, 4, 31]

1.3.1 Low Temperature Metal-Water Reactors

The aluminium-water reaction pathway for hydrogen generation has garnered significant international research attention, with most approaches focusing on nano-sized aluminium powder combined with catalysts at low temperatures. This approach involves reducing the aluminium powder to nano-sized particles and employing various catalytic methods such as alkaline solutions and ball milling the aluminium powder with other metals such as lithium and gallium, or salts such as NaCl. The primary aim of these methods is to increase the exposed surface area of the aluminium particles and prevent the protective layer of aluminium oxide from stopping or slowing down the reaction [32, 31]. This enables the metal-water reaction to take place below 120°C. Companies such as Found Energy are working to facilitate the metal-water reaction at room temperature and pressure [33].

This metal-water reactor design comes with both advantages and disadvantages. One advantage is that the reaction takes place quickly and enables the system to rapidly increase or decrease energy output to meet shifting load demands. However, this design also presents significant disadvantages:

- **High fuel production costs**: The production of nano-sized particles that also need to be ball milled with catalyst metals or salts requires an entire production chain and is likely to increase fuel costs substantially [34].

- **Elevated transport and handling costs**: The high reactivity of the nano-sized particles with water is likely to make transport and handling costs very high, as the metal reacts with water to produce hydrogen, which is a highly combustible gas and can independently contribute to climate change [35]. The higher production and handling costs for the fuel are likely to damage the economic case for this technology, since fuel costs constitute a large percentage of the operating costs for marine shipping, which is also a low-margin industry.

- **Low-grade heat output and reduced energy density**: As discussed in the previous sections, approximately half of the chemical potential energy of the aluminium metal is released as heat when reacted with water (as shown in equations 1 and 2). To maximise the round-trip efficiency of the metal-water reactor, this heat energy must be utilised for engine power. However, in these low-temperature reactors, the heat output is low-grade and unsuitable for use in traditional and high-efficiency steam engines [32]. The low-grade heat could be used by Stirling engines, but these engines have low power-to-weight ratios and have not been mass-produced so far like other heat engines such as diesel or gas turbines have [36]. The low-grade heat could also perhaps be upgraded using heat pumps and used in conjunction with Organic Rankine Cycle engines, but this increases the system complexity and the technology risks for the entire system [37]. The inability to utilise the low-grade heat also reduces the effective mass and volumetric energy density of the metal fuel by half.

Although the low-grade heat cannot be used for engine power, it can be used for other heating applications such as hot water production or indoor spas [38]. These applications are particularly useful in the yacht and cruise ship markets.

1.3.2 High Temperature Metal-Water Reactor

High-temperature metal-water reactors operate at 250–400°C and 15–25 MPa for optimal reaction rates [34]. These reactors can use micron-sized particles instead of nano-sized particles as in the case of low-temperature metal-water reactors. Although high-temperature metal-water reactors have garnered less research focus relative to low-temperature reactors, they come with some significant advantages, especially relating to marine shipping applications:

- **Lower fuel production costs**: Micron-sized particles are much cheaper and simpler to produce industrially compared to nano-sized particles [34, 4, 32].

- **Improved fuel handling and transport**: The micron-sized particles are also inert at room temperature and pressure, as opposed to nano-sized particles which can readily react with water at room temperature (depending on the catalytic process that is being used). Because of the fuel's inert nature, it makes them easier to transport and handle, which further improves the economic case for these reactors.

- **High-grade heat output and improved energy efficiency**: High-temperature reactors produce high-grade heat output in the form of a steam and hydrogen mixture. This high-grade heat can be used to power high-efficiency steam engines, which nearly doubles the round-trip energy efficiency of aluminium as a secondary energy carrier [32]. It also increases the mass and volumetric energy densities of the aluminium fuel as more useful energy can be created using the same kilogram of fuel.

Fuel Type	Fuel Mass Density (kWh/kg)	Engine Efficiency	By-Product to Fuel Weight Ratio	Final Energy Density (kWh/kg)
Al powder (ref.)	8.61	0.40 (ass.)	x	3.44
$Al(OH)_3$ by-product	8.61	0.40 (ass.)	2.89	1.19
AlOOH by-product	8.61	0.40 (ass.)	2.22	1.55
Al_2O_3 by-product	8.61	0.40 (ass.)	1.89	1.82

Table 1: Comparison of different $Al-H_2O$ by-product scenarios. Adapted from [2, 3, 4, 5, 6, 1]

Fuel Type	Volume Density (kg/L)	Final Volumetric Energy Density (kWh/L)
Al powder (ref.)	2.7	9.30
$Al(OH)_3$ by-product	2.42	2.88
AlOOH by-product	3.03	4.70
Al_2O_3 by-product	3.97	7.23

Table 2: Volumetric Energy Densities of different $Al-H_2O$ combustion scenarios. Adapted from [2, 3, 7, 8, 9, 10, 11]

- **Superior metal oxide by-product**: The aluminium-water reaction can produce three main by-products: $Al(OH)_3$, AlOOH, and Al_2O_3. The by-product produced can change the effective mass energy density of the fuel system. The lowest energy density happens with $Al(OH)_3$, while the highest is with Al_2O_3, as shown in Tables 1 and 2. In Fig 11 we can see that as temperature and pressure increase, the probability of producing Al_2O_3 instead of $Al(OH)_3$ increases, thus further increasing the effective mass and volumetric energy density of the aluminium fuel system. Al_2O_3 can also be used directly as the input in the Hall-Héroult process, as opposed to $Al(OH)_3$, which needs to be processed into Al_2O_3 first [38].

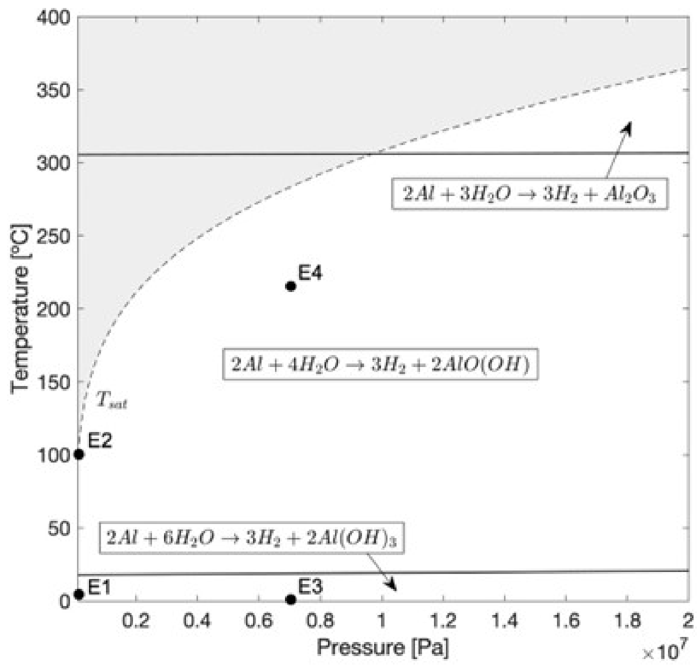

Figure 11: Al$-H_2O$ reaction transition diagram extrapolated to 10 MPa. Labels E1-E4 show conditions used for experimental validation [18]

However, high-temperature reactors come with their own disadvantages:

- **Extended reactor start-up time**: It could take some time for the reactor to heat up, which can be up to 15–30 minutes depending on the size of the reactor. This problem can be mitigated by the use of hybrid electric propulsion using electrochemical batteries that work as a buffer storage system as the reactor heats up. Hybrid electric systems can mitigate the impact on performance and manoeuvrability caused by the longer start-up time. These hybrid electric systems have been widely used in the automotive sector for decades and have already been gaining traction in the marine shipping sector [39]. Hybrid electric designs were also used to manage the longer start-up time for aluminium powder powered submarine and UUV systems that were researched by the American military [30].

- **Higher reactor manufacturing costs**: The reactor for the high-temperature system is likely to be more expensive since it needs to handle the higher temperature and pressure.

1.3.3 Summary Tables

Parameter	LT-MWR	HT-MWR
Particle Size	Nano-sized particles	Micron-sized particles
Catalysts	Range of special alloys (Li, Ga, Hg, etc.)	No catalysts
Temperature and Pressure	Below 120°C (can operate at room temperature depending on catalyst)	250–400°C and 15–25 MPa
Metal By-product	$Al(OH)_3$ or AlOOH	AlOOH or Al_2O_3
Worst Case Mass Energy Density	0.60 or 0.78 kWh/kg	1.55 or 1.82 kWh/kg
Worst Case Volumetric Energy Density	1.44 or 2.35 kWh/L	4.70 or 7.23 kWh/L
Thermal Heat Utilization	Difficult due to low temperature	Possible with simple steam turbines
Fuel Production Costs	Higher (smaller particles and ball milling alloys)	Lower (larger particles and low catalysts)
Round-trip Energy Eff.	11–13% (if no heat is utilized)	23–25%
Fuel Transport Costs	Higher (very reactive at room temp.)	Lower (mostly inert at room temperature)
Fuel Handling Safety	Lower	Higher
Reactor Costs	Lower	Higher (to withstand high temperature & pressure)
Reactor Start-up Time	Lower (ability to load follow)	15–30 minutes (better for baseloads)

Table 3: Comparison of Low Temperature Metal-Water Reactors (LT-MWR) and High Temperature Metal-Water Reactors (HT-MWR)

1.4 Economics of Aluminium Fuel

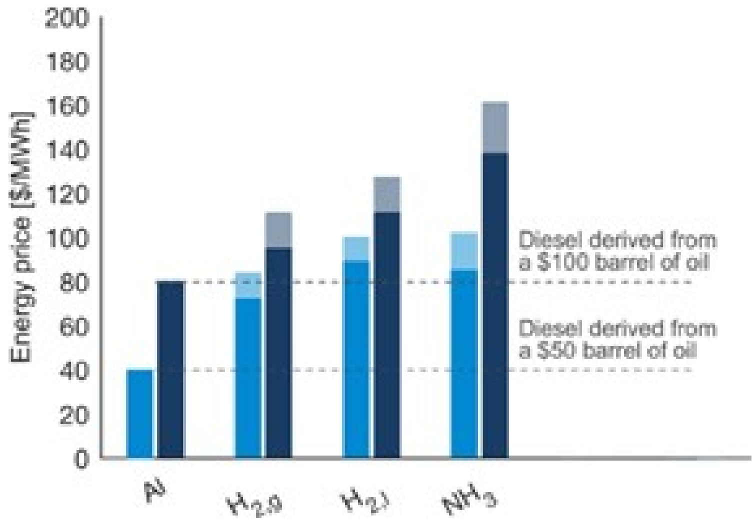

Figure 12: Aluminium fuel vs other electrofuels and diesel (The bars show the price of energy derived from low-carbon energy carriers, if the input electricity price were set to \$36/MWh (dark blue) or \$26/MWh (light blue). The pale region at the top of each bar represents the uncertainty related to the required inputs.) [1]

According to researchers at McGill University, Canada, if both the heat and hydrogen from the aluminium-water reaction are utilized for power generation at 40% efficiency, aluminium's round-trip efficiency as an electro-fuel reaches approximately 23-25%, making it comparable to hydrogen [40] and superior to ammonia [41].

The cost of electro-fuels like aluminium can be estimated based on electricity prices, while diesel is tied to crude oil prices. As shown in Figure 12, aluminium becomes cost-competitive with diesel when electricity prices fall between \$26-36/MWh and oil prices range from \$50-100 per barrel.

Aluminium also holds a cost advantage over rival electro-fuels such as hydrogen (both gas and liquefied forms) and ammonia. In OECD countries, industrial electricity rates range from \$45-143/MWh, with an average of \$100/MWh [42]. In regions with abundant renewable energy, such as Norway and Quebec, where rates are as low as \$45/MWh and \$27/MWh, aluminium becomes even more competitive in these markets [43].

1.5 Technology Development Pathway

The maritime shipping industry has historically been conservative in its adoption of new propulsion technologies. This reluctance stems from the sector's high capital costs and persistently low operating margins, which leave little room for experimental or unproven systems. Consequently, any novel engine or fuel system must first obtain certification from the International Maritime Organization (IMO), a process that typically requires extensive operational data and can take up to a decade. This lengthy and expensive approval pathway presents a formidable barrier to the introduction of a completely new fuel–engine combination, even in scenarios where high carbon taxes or stringent emissions regulations might otherwise favour low-carbon alternatives.

Given these challenges, a direct assault on the container shipping market with aluminium-water propulsion is unlikely to succeed in the near term. Moreover, while solar and battery costs are falling sharply, a robust green aluminium production industry has yet to emerge. Realistic projections suggest that at least 10–15 years will be required for this supply chain to mature and for aluminium fuel to reach cost parity with heavy fuel oil (HFO) and other hydrocarbons. Therefore, a beachhead market must be identified—one in which the technology can be developed, demonstrated, and refined through a "learning by doing" process, while generating the operational track record needed for eventual IMO certification.

Luxury Yachts and Superyachts as an Ideal Beachhead

The most promising initial market for aluminium-water propulsion systems is the luxury yacht and superyacht segment. These vessels are typically purchased by Ultra-High-Net-Worth Individuals (UHNWIs), defined as individuals with a net worth of US$30 million or more. The global UHNWI population is expected to grow by approximately 30% between 2025 and 2030, driving strong demand for bespoke, high-end vessels.

Unlike container shipping, which is a commoditised market driven almost exclusively by cost per tonne-mile, the superyacht market places a premium on product differentiation, innovation, and environmental stewardship. UHNWIs are also more likely to tolerate a significant green premium during the early years of technology development, while the aluminium-water propulsion system is being refined and while green aluminium fuel has yet to reach cost parity with conventional fuels. This tolerance for higher upfront costs provides a critical financial bridge for the technology.

Indeed, several green yachting options already exist, including fully battery-electric yachts powered by integrated solar panels (as shown in Fig. X). Furthermore, Austal—Australia's largest shipbuilder—is developing a catamaran ferry called the Gotland Horizon X, which features a lightweight "green aluminium" hull and can be powered entirely by green hydrogen. These developments indicate a strong and growing appetite for low-carbon, novel vessel designs within the premium marine segment.

Regulatory Pathway and Certification Advantages

Equally important are the regulatory differences between commercial shipping and the superyacht sector. IMO certification for container shipping follows a rigid, top-down process that demands extensive pre-approved data. In contrast, superyacht certification is handled primarily by private classification societies such as Registro Italiano Navale (RINA), Lloyd's Register (LR), and DNV. These societies employ a flexible, bottom-up, risk-based approach, often engaging in bespoke negotiations with builders and owners. They are specifically equipped to approve novel systems and one-off builds, making them far more accommodating to innovative propulsion technologies.

Pathway to Broader Adoption

If aluminium-water combustion systems can be successfully commercialised in the luxury yacht market, the operational data generated will greatly simplify the subsequent IMO certification process for container shipping applications. This strategy mirrors the path taken by Tesla in the automotive industry: the original Tesla Roadster—a high-performance, low-volume sports car—served as a beachhead to develop battery and electric drivetrain technology, generate real-world data, and build brand credibility before moving into mass-market vehicles. By targeting yachts and superyachts as the initial product development segment, the aluminium fuel system can follow an analogous learning-by-doing trajectory, ultimately paving the way for large-scale deployment in deep-sea shipping.

2 Research Questions

3 Methodology

4 Proposed Project Timeline

5 References

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